

5G RAN

eXn Self-Management Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for eXn self-management by the UMPThb board. For details, see: • 4.3.3 Hardware • 5.3.3 Hardware	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FOFD-010080	IPsec	5 Direct IPsec for eXn Interface

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
eXn self-management	None	4 eXn Self-Management
Direct IPsec for eXn interface	None	5 Direct IPsec for eXn Interface

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

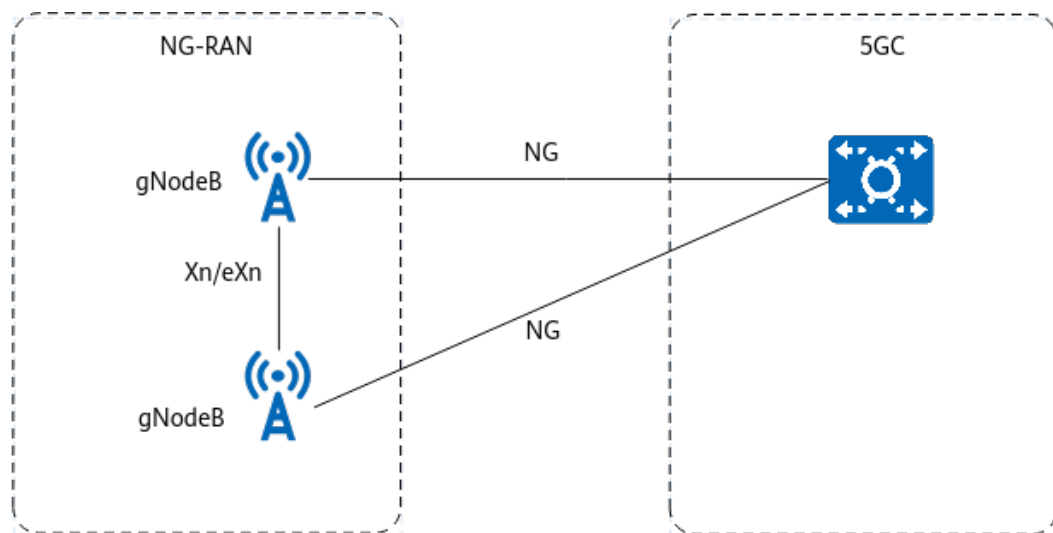
Function Name	Difference	Chapter/Section
eXn self-management	None	4 eXn Self-Management
Direct IPsec for eXn interface	None	5 Direct IPsec for eXn Interface

3 Overview

The eXn interface, similar to the Xn interface, is a logical interface between gNodeBs. eXn and Xn interfaces can coexist. The difference is that the eXn interface handles signaling and data only for inter-gNodeB service coordination, for example, inter-gNodeB carrier aggregation (CA); whereas, the Xn interface transfers signaling and data for basic inter-gNodeB services, for example, inter-gNodeB handovers.

Figure 3-1 shows the logical architecture of the NG, Xn, and eXn interfaces in SA networking.

Figure 3-1 Logical architecture of interfaces in SA networking



Service coordination-based features include:

- Inter-base-station carrier aggregation (CA)
- Coordinated interference management (low-frequency TDD)

eXn interfaces are classified into eXn-C and eXn-U interfaces. The eXn-C interface connects the control plane between gNodeBs, and the eXn-U interface connects the user plane between gNodeBs.

 NOTE

In the current version, eXn and Xn interfaces share the control plane. That is, the eXn control-plane information does not need to be configured separately, and the eXn user plane setup depends on the Xn control plane.

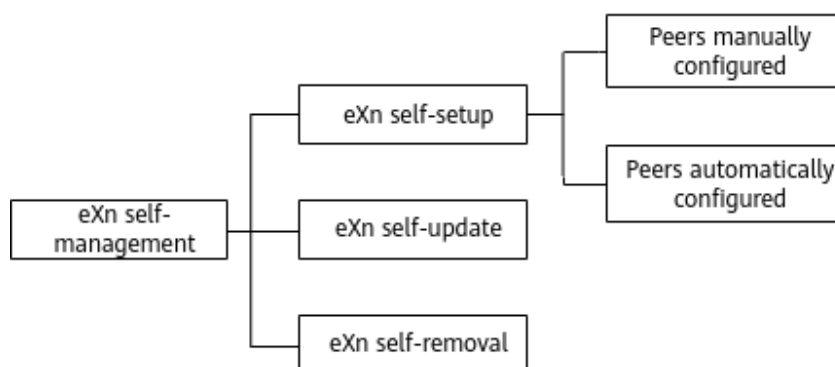
eXn self-management is introduced to simplify configuration operations and reduce operating expense (OPEX) for telecom operators. This function includes eXn self-setup, self-update, and self-removal.

4 eXn Self-Management

4.1 Principles

Figure 4-1 shows eXn self-management.

Figure 4-1 eXn self-management



If inter-gNodeB coordination services are required over the eXn interface, the **COORD_OUTPUT_BACKHAUL_PT** option of the **BBP.EXFUNC~~SW~~** parameter must be selected to enable transmission of coordination data over a backhaul port.

4.1.1 eXn Self-Setup

eXn self-setup is performed in either of the following methods:

- eXn self-setup with peers manually configured
USERPLANEPEER MOs are configured manually. If multiple eXn interface peers exist, multiple peer configurations must be manually added.
- eXn self-setup with peers automatically configured
Peers are configured automatically. When a UE initiates inter-gNodeB coordination services and the local gNodeB does not detect any eXn links, the local gNodeB obtains the peer gNodeB's user-plane IP address through the Xn link and automatically configures a peer for the eXn interface using the **USERPLANEPEER** MO.

In addition, packet filtering protected eXn self-setup can be configured for gNodeBs with packet filtering enabled.

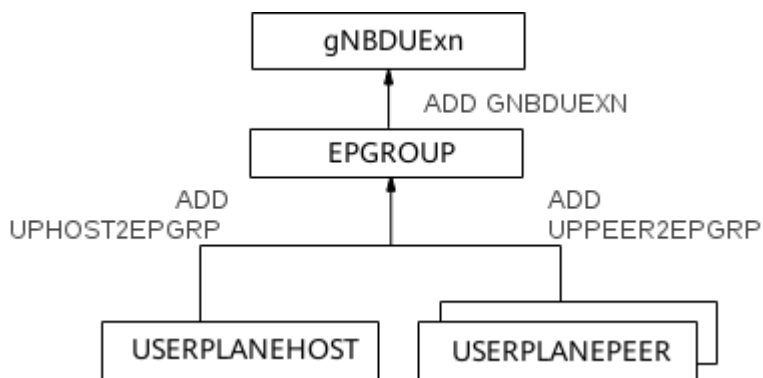
NOTE

In the current version, eXn self-setup requires an available Xn interface between the local and peer gNodeBs.

4.1.1.1 eXn Self-Setup with Peers Manually Configured

eXn interfaces are classified into eXn-C and eXn-U interfaces according to the data to be transmitted. In the current version, eXn and Xn interfaces share the control plane, and only eXn-U interfaces need to be configured. [Figure 4-2](#) shows how MOs for eXn self-setup relate to each other when peers are manually configured.

Figure 4-2 Relationships between MOs when peers are manually configured



- User plane

The **USERPLANEHOST** and **USERPLANEPEER** MOs are used to configure information such as the local and peer IP addresses of a user-plane link.

 - In IPv4 transmission, the gNodeB uses a local user-plane IPv4 address (specified by **USERPLANEHOST.LOCIPV4**) and a peer user-plane IPv4 address (specified by **USERPLANEPEER.PEERIPV4**).
 - In IPv6 transmission, the gNodeB uses a local user-plane IPv6 address (specified by **USERPLANEHOST.LOCIPV6**) and a peer user-plane IPv6 address (specified by **USERPLANEPEER.PEERIPV6**).
- Endpoint group

An **EPGROUP** MO is used to add local and peer user planes to an endpoint group.

 - The local and peer user planes of an eXn interface must be added to the same endpoint group.
 - A maximum of one IPv4-compliant **USERPLANEHOST** MO and one IPv6-compliant **USERPLANEHOST** MO can be simultaneously added to an **EPGROUP** MO. Multiple **USERPLANEPEER** MOs can be added to an **EPGROUP** MO.
- eXn interface

A **gNBDUExn** MO is used to add eXn interface objects and is bound to the **EPGROUP** MO.

 NOTE

When the eXn and Xn interfaces use the same local and peer user-plane IP addresses:

- The eXn and Xn interfaces must be configured with the same **EPGROUP** MO for their user planes. Otherwise, only the user-plane path in the **EPGROUP** MO in which the **USERPLANEPEER** MO is configured first can be set up, and the user-plane path in another **EPGROUP** MO cannot be set up.
- If the newly configured **EPGROUP** MO indicating the endpoint group in which the user-plane path is to be set up has the same local and peer user-plane IP addresses as an existing **EPGROUP** MO, delete the newly configured MO or ensure that the existing **EPGROUP** MO is referenced. Otherwise, the user-plane path setup will fail.

After the preceding configurations are complete, the gNodeB sets up an eXn-U transmission link based on the **USERPLANEHOST**, **USERPLANEPEER**, **gNBUEXn**, **EPGROUP**, and **TRANSALGEXTPARA** MOs.

Coordination-based features require a short eXn-U transmission link delay. Therefore, it is recommended that the delay be queried and evaluated using the following methods before and after coordination-based features are enabled:

- Before coordination-based features are enabled: (1) You can activate TWAMP in light architecture. (Activate the TWAMP Responder function first and then the TWAMP Controller function. For details, see *IP Active Performance Measurement*.) Then, run the **DSP TWAMPSENDER** command to query the eXn-U transmission link delay. (2) Alternatively, you can manually configure an IP performance monitoring (IP PM) session that uses uplink activation, to measure the eXn-U transmission link delay. For details, see *IP Performance Monitor*. Then, run the **DSP IPPMSESSION** command to query the eXn-U transmission link delay.
- Before coordination-based features are enabled, it is good practice to select the **COORD_IPPM_DSCP_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter to specify the priority of IP PM packets for coordination-based services.
 - If this option is deselected, DSCP 48 is used for the IP PM sessions automatically established for eXn coordination-based services.
 - If this option is selected, DSCP 0 is used for the IP PM sessions automatically established for eXn coordination-based services. This reduces the priority of probe packets for such IP PM sessions to the same level as actual data packets for coordination-based services. In this way, the congestion check based on IP PM probe packets can produce more accurate results.
- After coordination-based features are enabled, the gNodeB automatically creates an IP PM session to measure the eXn-U transmission link delay. In this case, you can run the **DSP GNBUEXNUPINFO** command to query for the delay.

NOTE

IP PM sessions can be manually configured only when uplink activation is configured, and cannot be manually configured when bidirectional activation is configured. For details about how to configure IP PM sessions, see *IP Performance Monitor*.

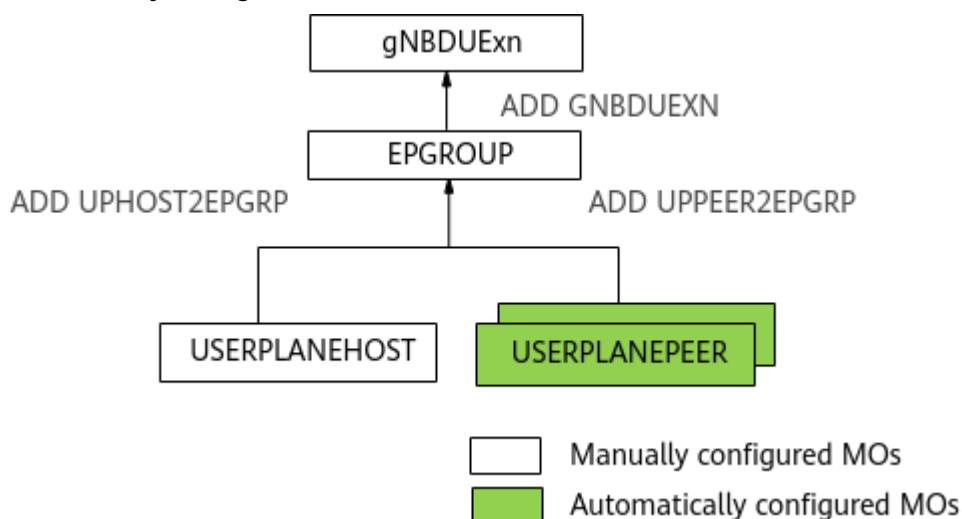
After coordination-based features are enabled, it is recommended that IP PM sessions manually configured for the eXn-U transmission links be deactivated. Otherwise, performance of coordination-based services may be affected.

Operator-specific eXn interface configuration is supported in RAN sharing with dedicated carrier. For details, see *Multi-Operator Sharing*.

4.1.1.2 eXn Self-Setup with Peers Automatically Configured

Figure 4-3 shows how MOs for eXn self-setup relate to each other when **USERPLANEPEER** MOs are automatically configured.

Figure 4-3 Relationships between MOs for self-setup when peers are automatically configured



Self-setup with peers automatically configured is recommended because it reduces manual operations, as well as network planning and optimization costs.

This method requires the following conditions to be met:

- There is an available Xn interface between gNodeBs.
- eXn-related MOs, such as **USERPLANEHOST**, **gNBDUExn**, and **EPGROUP**, have been configured.
- A UE initiates inter-base-station coordination services.

In this method, when a UE initiates inter-gNodeB coordination services and the local gNodeB does not detect any eXn links, the local gNodeB obtains the peer gNodeB information automatically only through the Xn-C interface. Then, it configures the **USERPLANEPEER** MO to perform eXn self-setup.

4.1.1.3 Packet Filtering Protected eXn Self-Setup

If integrated firewall is enabled and packet filtering is configured on a gNodeB, access control list (ACL) rules in the ACL associated with the packet filtering (in

the **ACLRULE** MO under the **ACL** MO for IPv4, or in the **ACLRULE6** MO under the **ACL6** MO for IPv6) must be configured for automatically established eXn-U paths. ACL rules can be manually or automatically configured.

- If **EPGROUP.PACKETFILTERSWITCH** (IPv4)/**EPGROUP.PACKETFILTERSWITCH6** (IPv6) is set to **ENABLE** for an eXn object (**gNBDUExn** MO), the ACL rule self-configuration function for packet filtering is enabled during eXn self-setup. That is, ACL rules are automatically created for the eXn-U interface during eXn self-setup based on the local and peer configurations of the eXn interface. The ACL rules are then automatically added to the ACL for packet filtering.
 - You are not advised to manually modify the automatically created packet filtering rules. If you manually modify the automatically generated packet filtering rules, the eXn-U links may be disconnected. In this case, you can restore the modified packet filtering rules or delete the incorrect packet filtering rules. Packet filtering rules will be automatically created in the next eXn self-setup procedure.
 - If the gNodeB determines that the number of ACL rules has exceeded the maximum allowed number, MML command output will indicate that the number exceeds board capability. In this case, it is recommended that ACL rules be manually configured.
- If **EPGROUP.PACKETFILTERSWITCH** (IPv4)/**EPGROUP.PACKETFILTERSWITCH6** (IPv6) is set to **DISABLE** for an eXn object (**gNBDUExn** MO), ACL rules must be manually configured on the gNodeB for the peer base station.

 NOTE

- When the **GTRANSPARA.TRANSFCFGMODE** parameter is set to **OLD**, the old model is used for transmission configuration and packet filtering is configured in the **PACKETFILTER** MO. When the **GTRANSPARA.TRANSFCFGMODE** parameter is set to **NEW**, the new model is used for transmission configuration and packet filtering is configured in the **PACKETFILTERING** MO.
- For details about packet filtering and ACL rule configuration, see *Equipment Security*.

4.1.2 eXn Self-Update

eXn self-update refers to eXn peer information self-update, which can be service-triggered or manually triggered.

4.1.2.1 Service-Triggered Self-Update of eXn Peer Information

If the peer user-plane IP address changes, the eXn interface to the peer end will become faulty. During eXn self-setup, the local gNodeB obtains the new peer user-plane IP address, and generates a new **USERPLANEPEER** MO based on the new IP address. The faulty user-plane path is removed through eXn-U self-removal. For details, see [4.1.3 eXn Self-Removal](#).

4.1.2.2 Manually Triggered Self-Update of eXn Peer Information

The local IP address of the eXn interface is referenced by the **USERPLANEPEER** MO of the peer end. During the change of the local IP address, if the local and peer gNodeBs are managed by the same MAE-Deployment, the MAE-Deployment

can automatically associate and change the local IP address referenced by the peer end in the same virtual routing and forwarding (VRF) domain.

4.1.3 eXn Self-Removal

The gNodeB can automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **AUTO_MODE** and the corresponding eXn-U transmission links. The gNodeB does not automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **MANUAL_MODE** and the corresponding eXn-U transmission links. You can manually remove those links by referring to [Deactivation Command Examples](#). eXn self-removal is controlled by the **GEPMODELpara.UPAUTODELSW** parameter.

After eXn self-removal is enabled, the gNodeB determines whether to perform eXn-U self-removal based on one of the following mechanisms:

- Self-removal based on link faults

It is recommended that the GTP-U static detection function be enabled (by setting the **GTPU.STATICCHK** parameter to **ENABLE**) to detect faulty eXn-U transmission links in a prompt manner. If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.
- Self-removal based on the aging mechanism
 - In case of an excess over the specifications of a base station

The gNodeB marks a transmission link as aged if the link is not used by any service.

 - The gNodeB performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
 - When UEs need to establish new transmission links for initiating services, the gNodeB checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
 - If the number of established transmission links has reached the maximum number supported and there is no aged link when UEs need to establish new transmission links for initiating services, the gNodeB rejects new UEs' access requests.
 - In case of an excess over the specifications of a single endpoint group

"Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the gNodeB when the

SINGLE_EPUP_AUTO_DEL_SW option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter is selected. The base station removes aged links when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed. The condition for marking a link as aged and the self-removal mechanism are the same as those for "self-removal in case of an excess over the specifications of a base station".

4.1.4 Restrictions on the eXn Interface

Application Restrictions

- eXn self-setup requires an available Xn interface between the local and peer gNodeBs.
- An eXn interface supports one user-plane local IPv4 address and one user-plane local IPv6 address at the same time.
- The protocol version configured for the eXn interface must be the same as that configured for the Xn interface of the corresponding operator. For example, the protocol version for the eXn interface must be IPv4 if that is IPv4 for the Xn interface, IPv6 if that is IPv6 for the Xn interface, or IPv4+IPv6 dual stack if that is IPv4+IPv6 dual stack for the Xn interface. In RAN sharing scenarios, if operators use different Xn configurations (for example, the Xn interface is configured to use IPv4 for one operator and IPv6 for another, or IPv4+IPv6 dual stack for one operator and single stack for another), eXn setup across base stations is not supported.
- The same IP version must be used for both the user-plane path of the eXn interface and the control-plane link of the Xn interface. That is, the **EPGROUP.IPVERPREFERENCE** parameter needs to be set to the same value for the user plane of the eXn interface and the control plane of the Xn interface. The **EPGROUP.IPVERPREFERENCE** parameter must be set to the same value for the local and peer gNodeBs over an eXn interface.
- The eXn interface does not support BBU interconnection between base stations.
- Only one **gNBDUExn** MO can be configured for a gNodeB and multiple operators can share the eXn interface. If operator-specific eXn interface configuration is performed in RAN sharing with dedicated carrier, the **gNBDUExn** MO (if any) needs to be removed before configuring operator-specific **gNBDUExnOp** MOs. If no **gNBDUExn** MO exists, operator-specific **gNBDUExnOp** MOs are directly configured. For details, see *Multi-Operator Sharing*.
- The eXn interface does not support the simplified endpoint mode.
- The eXn interface does not support "Measurement of UserPlaneHostIP Performance" and "Transport Auto Setup User Plane Monitoring" functions.
- The end-to-end maximum transmission unit (MTU) of eXn transmission links must be greater than or equal to 1500 bytes in both secure and non-secure networking. Otherwise, coordination packet fragmentation may occur, affecting coordination services.

4.2 Network Analysis

4.2.1 Benefits

eXn self-management provides the following benefits:

- Simplifies configuration operations, reduces the OPEX of operators, and improves gNodeB resource usage.
- Enables established eXn interfaces to provide inter-gNodeB service coordination-based features, which can improve user experience or spectral efficiency. Coordination services are sensitive to delay. Network out-of-order and fragmentation cause extra delay. As a result, coordination service gains are affected. You are advised to check the transport network before configuring the eXn interface to prevent out-of-order and fragmentation on transmission devices.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

After the **COORD_IPPM_DSCP_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter is selected, the network impacts are as follows:

- For CA services over the eXn interface, when eXn flow control on received traffic is triggered, the volume of split traffic is reduced to prevent service loss caused by congestion-induced packet loss.
- For CoMP services over the eXn interface, when eXn flow control on received traffic is triggered, coordination is suspended to prevent service loss caused by congestion-induced packet loss.
- When the service load over the eXn interface is heavy, for UEs that are served by different BBUs:
 - **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** may increase.
 - The values of the following counters related to the uplink and downlink IBLERs may decrease:
 - **N.DL.SCH.QPSK.ErrTB.Ibler**
 - **N.DL.SCH.16QAM.ErrTB.Ibler**
 - **N.DL.SCH.64QAM.ErrTB.Ibler**
 - **N.DL.SCH.256QAM.ErrTB.Ibler**
 - **N.UL.SCH.QPSK.ErrTB.Ibler**
 - **N.UL.SCH.16QAM.ErrTB.Ibler**
 - **N.UL.SCH.64QAM.ErrTB.Ibler**
 - **N.UL.SCH.256QAM.ErrTB.Ibler**
 - The values of the following counters related to the uplink and downlink RBLERs may decrease:

- N.DL.SCH.QPSK.ErrTB.Rbler
- N.DL.SCH.16QAM.ErrTB.Rbler
- N.DL.SCH.64QAM.ErrTB.Rbler
- N.DL.SCH.256QAM.ErrTB.Rbler
- N.UL.SCH.QPSK.ErrTB.Rbler
- N.UL.SCH.16QAM.ErrTB.Rbler

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Automatic user plane removal	GEPMODEL PARA.UPAUT ODELSW	<i>eXn Self- Management</i>	Ensure that the GEPMODEL PARA.UPAUTODELSW parameter is set to ENABLE before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA.TNLBEARER MNGALGSW parameter).
Low-frequency TDD High-frequency TDD	Self-removal in case of an excess over the specifications of a single endpoint group	SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA.TNLBEARER MNGALGSW parameter	<i>eXn Self- Management</i>	Ensure that "self-removal in case of an excess over the specifications of a single endpoint group" is disabled before disabling automatic user plane removal (by setting the GEPMODEL PARA.UPAUTODELSW parameter to DISABLE).

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Coordination over ideal backhaul	None	<i>Cloud BB Overview and USU3910-based Multi-BBU Interconnection in eRAN Feature Documentation</i>	In LTE and NR co-BBP scenarios, when the COORD_OUTPUT_BA CKHAUL_PT option of the BBP.EXFUNCSW parameter is selected, coordination over ideal backhaul cannot be configured for HEI port interconnection.
Low-frequency TDD High-frequency TDD	Common clock through HEI port interconnection	None	<i>Common Clock</i>	When the COORD_OUTPUT_BA CKHAUL_PT option of the BBP.EXFUNCSW parameter is selected, HEI ports on BBPs cannot be interconnected to share clock signals.
Low-frequency TDD High-frequency TDD	User plane endpoint group ID for the NG interface	gNBCUNg.UpEpGroupId	<i>NG and Xn Self-Management</i>	The value of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface must be different from that of the gNBCUNg.UpEpGroupId parameter configured for the NG interface.
Low-frequency TDD High-frequency TDD	User plane endpoint group ID for the S1 interface	gNBCUS1.UpEpGroupId	<i>X2 and S1 Self-Management in NSA Networking</i>	The value of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface must be different from that of the gNBCUS1.UpEpGroupId parameter configured for the S1 interface.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Endpoint group IDs for the X2 interface	gNBCUX2.CpEpGroupId and gNBCUX2.UpEpGroupId	<i>X2 and S1 Self-Management in NSA Networking</i>	If the value of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface is different from that of the gNBCUX2.UpEpGroupId parameter configured for the X2 interface, the value of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the eXn interface must be different from that of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the X2 interface.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	User plane endpoint group ID for the Xn interface	gNBCUXn.UpEpGroupId	<i>NG and Xn Self-Management</i>	- If the value of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface is different from that of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface, the value of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the eXn interface must be different from that of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the Xn interface. - If the gNBDUExn.UpEpGroupId parameter of the eXn interface is set to the same value as the gNBCUXn.UpEpGroupId parameter of the Xn interface, the number of USERPLANEHOST.UPHOSTID parameters referenced in the EPGROUP MO cannot exceed 2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Application type	EPGROUP.APPTYPE	None	The EPGROUP.APPTYPE parameter referenced by the eXn interface must be set to NULL .
Low-frequency TDD High-frequency TDD	Local end point group type flag	EPGROUP.TYPEFLAG	None	The EPGROUP.TYPEFLAG parameter referenced by the eXn interface must be set to COMMON .
Low-frequency TDD High-frequency TDD	User-plane host active/standby flag	USERPLANEHOST.FLAG	None	The USERPLANEHOST.FLAG parameter referenced by the eXn interface must be set to MASTER .

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Only the UBBPh board does not support coordination data transmission over the backhaul port. The UBBPh board supports inter-base-station coordination services by default. Therefore, coordination data transmission over the backhaul port does not need to be enabled.

The board requirements for the other functions are as follows:

Only the UMPThb, UMPTha, UMPTg, UMPTga, and UMPTe series main control boards, and only the UBBPg, UBBPh, and UBBPfw series baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Table 4-1 lists the maximum number of neighboring base stations that can be supported over the eXn (this number varies with the main control board).

Table 4-1 Maximum number of neighboring base stations that can be supported over the eXn

Main Control Board	Maximum Number of Neighboring Base Stations That Can Be Supported over the eXn
UMPTe	200
UMPTg	300
UMPTga	300
UMPT _{ha}	300
UMPT _{hb}	300

RF Modules

There are no requirements for RF modules.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

The following table describes the parameters that must be set in a **BBP** MO to configure the switch for transmitting coordination data over a backhaul port.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BBP.CN	Set this parameter based on the network plan.
Subrack No.	BBP.SRN	Set this parameter based on the network plan.
Slot No.	BBP.SN	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Base Band Work Standard	BBP.BBWS	The values of this parameter must include NR to enable eXn self-management.
Extended Function Switch	BBP.EXFUNC SW	Select the COORD_OUTPUT_BACKHAUL_P T option of the BBP.EXFUNC SW parameter to enable eXn self-management.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO to configure the user-plane host.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPH OSTID	Set this parameter based on the network plan.
IP Version	USERPLANEHOST.IPVER SION	Set the IP version of the local end to the same as that of the peer end.
Local IP Address	USERPLANEHOST.LOCIP V4	If USERPLANEHOST.IPVERSION is set to IPv4 , configure a local user-plane IPv4 address.
Local IPv6 Address	USERPLANEHOST.LOCIP V6	If USERPLANEHOST.IPVERSION is set to IPv6 , configure a local user-plane IPv6 address.

(Optional) The following table describes the parameters that must be set in a **USERPLANEPEER** MO. This MO is required only for manual peer configuration.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	USERPLANEPEER.UPPEE RID	Set this parameter based on the network plan.
IP Version	USERPLANEPEER.IPVER SION	Set the IP version of the local end to the same as that of the peer end.
Peer IP Address	USERPLANEPEER.PEERI PV4	If USERPLANEPEER.IPVERSION is set to IPv4 , configure a peer user-plane IPv4 address.

Parameter Name	Parameter ID	Setting Notes
Peer IPv6 Address	USERPLANEPEER.PEER/IPv6	If USERPLANEPEER.IPVERSION is set to IPv6 , configure a peer user-plane IPv6 address.

The following table describes the parameters that must be set in an **EPGROUP** MO. It is recommended that the **USERPLANHOST** and **USERPLANEPEER** MOs for each operator be added to the same **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
Packet Filter ACL Rule Auto-Setup-Deletion SW	EPGROUP.PACKETFILTE RSWITCH	Set this parameter based on the network plan.
IPv6 Packet Filter ACL Rule Auto-Setup-Deletion SW	EPGROUP.PACKETFILTE RSWITCH6	Set this parameter based on the network plan.
Static Check Switch	EPGROUP.STATICCHK	The value ENABLE is recommended.
IP PM Automatic Setup and Deletion Switch	EPGROUP.IPPMSWITCH	Set this parameter based on the network plan.
Application Type	EPGROUP.APPTYPE	Set this parameter to NULL .

The following table describes the parameters that must be set to configure the reference relationship between the **gNBDUExn** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
gNodeB DU eXn ID	gNBDUExn.gNBDuExnId	Set this parameter based on the network plan.
User Plane Endpoint Group ID	gNBDUExn.UpEpGroupI d	Set this parameter based on the network plan.

The following table describes the parameters that must be set in the **TRANSALGEXTPARA** MO to configure the priority of IP PM packets for coordination-based services.

Parameter Name	Parameter ID	Setting Notes
Transport Bearer Management Algo Switch	TRANSALGEXTPARA.TN LBEARERMNGALGSW	It is recommended that the COORD_IPPM_DSCP_SW option be selected.

If the SON function is to be used to automatically configure, update, and remove the eXn interface, the **GTRANSPARA.EPOBJAUTOCFGSW** parameter must use its default value (**ENABLE**).

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

When an eXn interface and an Xn interface use different user-plane IP addresses, the MML command examples for setting up the eXn interface are as follows:

```
//Turning on the switch for coordination data transmission over the backhaul port
MOD BBP: CN=0, SRN=0, SN=2, BBWS=NR-1, EXFUNCSW=COORD_OUTPUT_BACKHAUL_PT-1;
//Adding an endpoint group for the eXn interface (IPv4 used as an example)
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH=DISABLE, STATICCHK=ENABLE,
IPPMSWITCH=DISABLE, APPTYPE=NULL;
//Adding an eXn user-plane host
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;
//Optional Adding an eXn user-plane peer for manually configuring eXn user-plane's peer gNodeB
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="5.5.3.21", IPSECSWITCH=DISABLE;
//Adding the eXn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Optional Adding the eXn user-plane peer to the endpoint group for manually configuring eXn user-
plane's peer gNodeB
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding an eXn object on the gNodeB
ADD GNBDUEXN: gNBDuExnId=1, UpEpGroupId=1;
```

When an eXn interface and an Xn interface use the same user-plane IP address, the MML command examples for setting up the eXn interface are as follows:

```
//Turning on the switch for coordination data transmission over the backhaul port
MOD BBP: CN=0, SRN=0, SN=2, BBWS=NR-1, EXFUNCSW=COORD_OUTPUT_BACKHAUL_PT-1;
//Adding an endpoint group for the Xn/eXn interface (IPv4 used as an example)
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH=DISABLE, STATICCHK=ENABLE,
IPPMSWITCH=DISABLE, APPTYPE=NULL;
```

```
//Adding an Xn/eXn user-plane host
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;
//(Optional) Adding an Xn/eXn user-plane peer for manually configuring Xn/eXn user-plane's peer gNodeB
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="5.5.3.21", IPSECSWITCH=DISABLE;
//Adding the Xn/eXn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//(Optional) Adding the Xn/eXn user-plane peer to the endpoint group for manually configuring Xn/eXn
user-plane's peer gNodeB
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding a gNodeB CU Xn object
ADD GNBCUXN: gNBCuXnId=1, CpEpGroupId=2, UpEpGroupId=1;
//Adding an eXn object on the gNodeB
ADD GNBDUEXN: gNBDuExnId=1, UpEpGroupId=1;
```

Optimization Command Examples

The following is an example of configuring the priority of IP PM packets for coordination-based services:

```
//Configuring the priority of IP PM packets for coordination-based services
SET TRANSALGEXTPARA: TNLBEARERMNGALGSW=COORD_IPPM_DSCP_SW-1;
```

Deactivation Command Examples

The MML command examples for removing an eXn interface are as follows:

```
//Turning off the switch for coordination data transmission over the backhaul port
MOD BBP: CN=0, SRN=0, SN=2, BBWS=NR-1, EXFUNCSW=COORD_OUTPUT_BACKHAUL_PT-0;
//Removing a single eXn-U link and the USERPLANEPEER MO
//EPGROUPID is the UPEPGROUPID referenced by the eXn interface, and UPPEERID is the to-be-removed
user-plane peer ID of the eXn interface.
RMV UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
RMV USERPLANEPEER: UPPEERID=0;
//Removing the eXn object from the gNodeB
RMV GNBDUEXN: gNBDuExnId=1;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

eXn Self-Setup

- Step 1** Run the **DSP GNBCUXNINTERFACE** command to query for the Xn interface information. There is an available Xn link if the value of **gNodeB CU Xn Interface State** is **Normal** and the Xn link supports the PLMN selected by UEs.
- Step 2** Enable the gNodeB to initiate inter-gNodeB coordination services to trigger eXn setup.
- Step 3** Run the **DSP GNBDUEXNUPINFO** command to query for the information about the eXn-U path established on the gNodeB. If the value of **gNodeB DU eXn User Plane Link State** is **Normal**, the eXn interface has been set up.

----End

SON Logs

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** On the **Query SON Log** tab page, select **NR eXn Interface Self Management Log** from the **Log Category** drop-down list box. In the **Event Name** area, select **Automatically Establish Service Link**, **Automatically Delete Service Link**, and **Automatically Update eXn User Plane Link**. Then, select target base stations in the **Log Source** area to check different types of eXn self-configuration operation logs.

----End

Expected result: The SON logs contain eXn self-setup, self-removal, and self-update events.

4.4.3 Network Monitoring

If the eXn interface is faulty, check whether the alarm listed in [Table 4-2](#) has been reported. If an alarm has been reported, clear the alarm by referring to the alarm reference.

Table 4-2 Alarms related to the eXn interface

Alarm ID	Alarm Name	Alarm Severity
25954	User Plane Fault	Major

5 Direct IPsec for eXn Interface

5.1 Principles

This function creates direct IPsec tunnels between two gNodeBs to reduce transmission delay over an eXn interface when connected through a security gateway (SeGW), as routing through the SeGW can introduce unwanted delay, depending on its location and the network topology. When direct IPsec is configured on an eXn interface with gNodeBs at two ends (but in different network segments), respective routes between the two gNodeBs must be configured.

Figure 5-1 and **Figure 5-2** show the typical networking of direct IPsec for the eXn interface between gNodeBs.

Figure 5-1 Typical networking of direct IPsec (type 1)

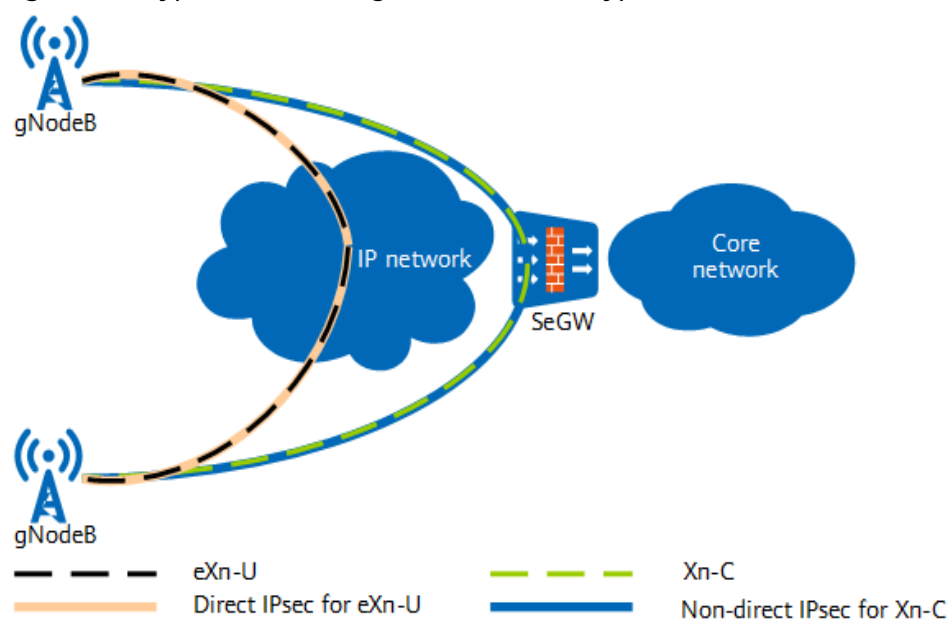
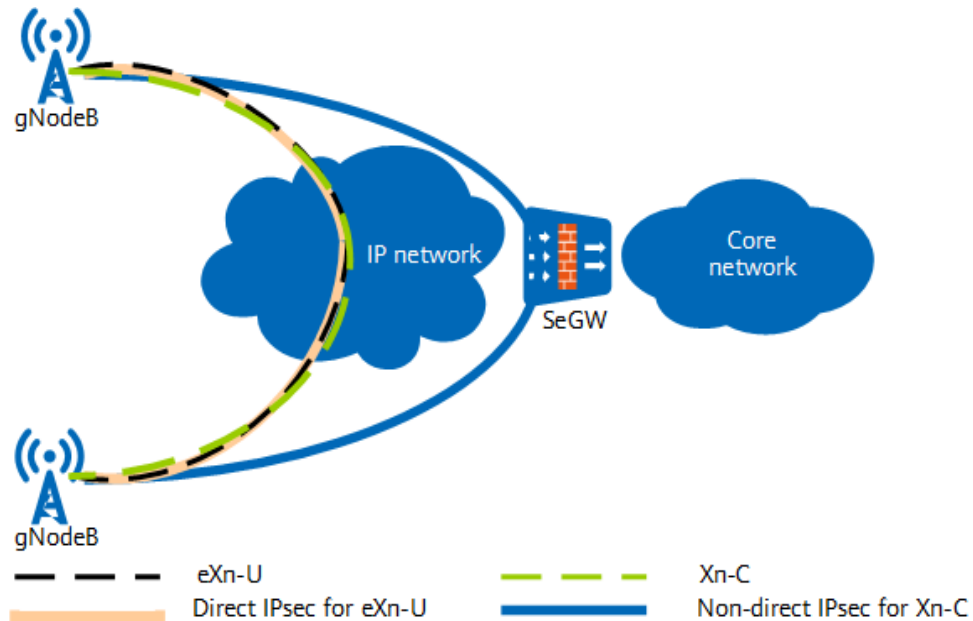


Figure 5-2 Typical networking of direct IPsec (type 2)



If eXn interfaces adopt IPsec tunnels passing through the SeGW (non-direct IPsec transmission paths) and the service delay requirements are not met, it is recommended that the transmission delay of direct IPsec transmission paths of the eXn interfaces be measured before direct IPsec is deployed:

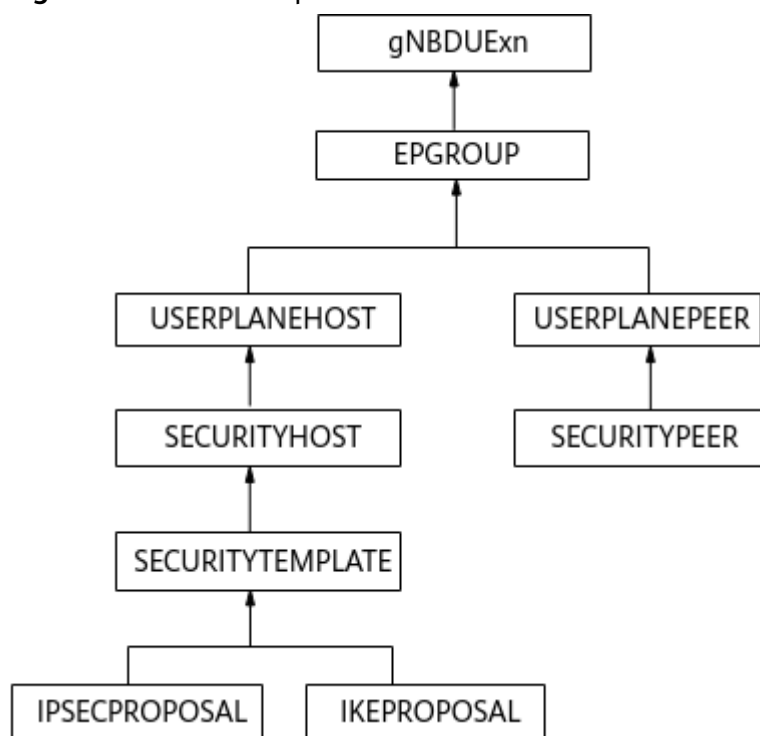
- If the measured delay is less than the transmission delay of non-direct IPsec transmission paths of the eXn interface and meets the delay requirements of to-be-deployed coordination-based features, the transport network performance allows for direct IPsec deployment.
- If the measured delay is greater than or equal to the transmission delay of non-direct IPsec transmission paths of the eXn interface or does not meet the delay requirements of to-be-deployed coordination-based features, the transport network performance does not allow for direct IPsec deployment. In this case, it is recommended that the transport network be optimized until the requirements for direct IPsec deployment are met.

In the current version, only the user plane of an eXn interface can be configured. Therefore, only direct IPsec for the eXn-U interface is supported.

5.1.1 Direct IPsec Setup

The eXn user-plane peer (specified by the **USERPLANEPEER** MO) can be automatically or manually configured to achieve direct IPsec setup between gNodeBs. Direct IPsec is classified into direct IPv4 IPsec and direct IPv6 IPsec. If the **USERPLANEPEER** MO is manually configured, the **SECURITYPEER** MO must be configured as well. [Figure 5-3](#) shows how MOs relate to each other.

Figure 5-3 Relationships between MOs



The following describes how the MOs in [Figure 5-3](#) relate to each other.

- User plane

The **USERPLANEHOST** and **USERPLANEPEER** MOs are used to configure information such as the local and peer IP addresses of a user-plane path. The **SECURITYHOST** and **SECURITYPEER** MOs are used to configure information such as the local and peer security IP addresses of the user-plane path.

- In IPv4 networking, the gNodeB uses a local eXn user-plane security IPv4 address (specified by **SECURITYHOST.LOCIKEIPV4**) and a peer eXn user-plane security IPv4 address (specified by **SECURITYPEER.PEERIKEIPV4**). Additionally, **USERPLANEHOST.IPSECSWITCH** must be set to **ENABLE**.
- In IPv6 networking, the gNodeB uses a local eXn user-plane security IPv6 address (specified by **SECURITYHOST.LOCIKEIPV6**) and a peer eXn user-plane security IPv6 address (specified by **SECURITYPEER.PEERIKEIPV6**). Additionally, **USERPLANEHOST.IPSECSWITCH** must be set to **ENABLE**.

- Security parameters

The **IPSECPROPOSAL**, **IKEPROPOSAL**, and **SECURITYTEMPLATE** MOs are used to add security-related configurations and bind them to user-plane information.

After the preceding MOs are configured, the gNodeB sets up eXn-U transmission links and IPv4/IPv6 IPsec tunnels based on the MOs.

NOTE

A peer IKE IP address (specified by **SECURITYPEER.PEERIKEIPV4** or **SECURITYPEER.PEERIKEIPV6**) can be used to set up only one IPv4/IPv6 IPsec tunnel.

In IPv4 networking, when an eXn interface is configured with both direct IPv4 IPsec tunnels and IPv4 IPsec tunnels passing through the SeGW, the

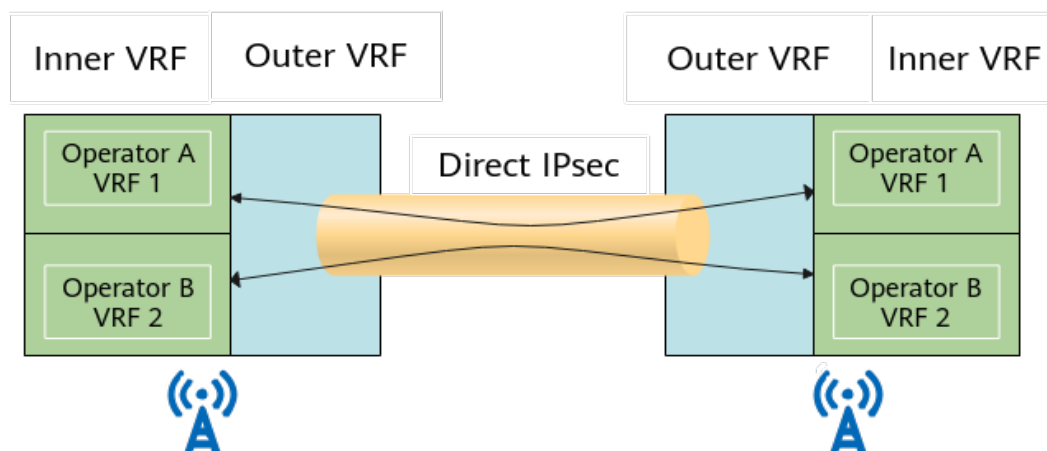
GTRANSPARA.DIRECTIPSECPRIMATCHSW parameter can be used to adjust the IPv4 IPsec policy matching priority. This parameter specifies whether the eXn interface preferentially matches either direct IPv4 IPsec tunnels or IPv4 IPsec tunnels passing through the SeGW.

In IPv6 networking, when an eXn interface is configured with both direct IPv6 IPsec tunnels and IPv6 IPsec tunnels passing through the SeGW, the eXn interface preferentially uses the direct IPv6 IPsec tunnels. This is because the **POLICYBASEDROUTING6** MO has a higher priority than the **IPROUTE6** and **SRCIPROUTE6** MOs.

5.1.2 Direct IPsec Sharing Among Multiple Operators

In multi-operator RAN sharing with dedicated carriers, a direct IPsec tunnel can be shared by the eXn interfaces of different operators using the direct IPsec demultiplexing technology. Without the need for additional direct IPsec tunnels, direct IPsec services are provided for multiple operators based on their VPN IDs. This saves resources and improves the usage rate of direct IPsec. Different inner VRFs are configured for different operators. IPsec can use VRFs to isolate outer packets from inner service packets. This function enables deployment of multiple operators on different inner VRFs of a base station. In this way, multiple operators can use a direct IPsec tunnel to share eXn services. In addition, eXn services no longer need to pass through the SeGW. [Figure 5-4](#) shows the principle using two operators as an example.

Figure 5-4 Direct IPsec sharing among multiple operators



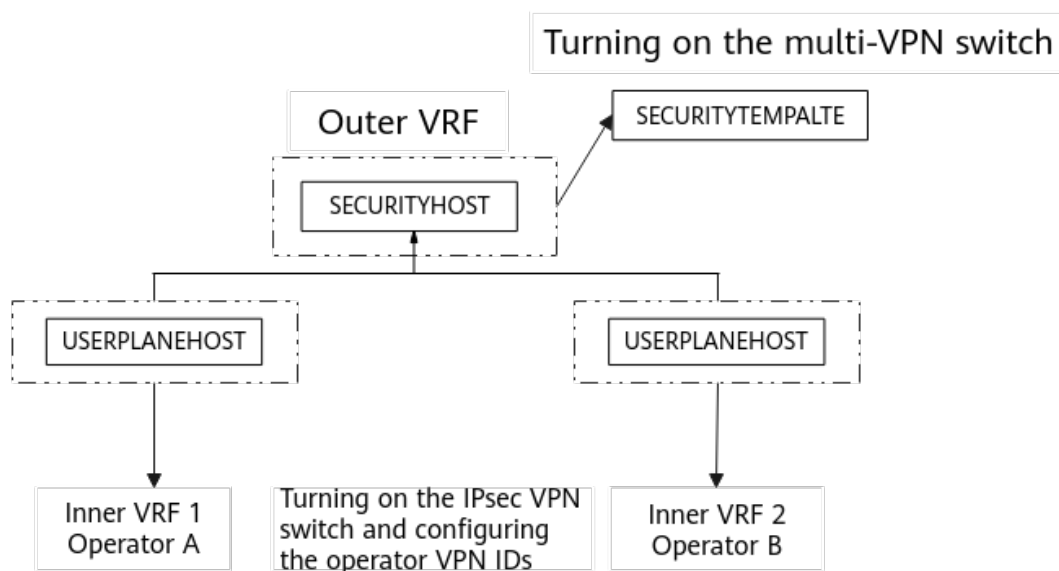
NOTE

- In RAN sharing with common carrier, multiple operators cannot deploy their own eXn interfaces. Therefore, this function is not involved in RAN sharing with common carrier.
- For details about RAN sharing with dedicated carriers and RAN sharing with common carrier, see *Multi-Operator Sharing*.

In multi-operator sharing scenarios, after the multi-VPN switch is turned on (by using the **SECURITYTEMPLATE.MULTIVPNSWITCH** parameter), the IPsec VPN switch is turned on (by using the **VRF.IPSECVPNSW** parameter) and the corresponding operator VPN ID is configured (by using the **VRF.IPSECVPNID** parameter) to implement direct IPsec sharing among multiple operators. The VPN IDs (specified by the **VRF.IPSECVPNID** parameter) must be unique for different

operators and the VPN ID of each operator must be consistent across all of its base stations. Otherwise, direct IPsec sharing among multiple operators cannot work properly. [Figure 5-5](#) shows the configuration model. After the multi-VPN switch is turned on, multiple operators (up to six) can reference the same direct IPsec tunnel. The multi-VPN switch referenced by the **SECURITYTEMPLATE** MO and the IPsec VPN switch referenced by the **SCTPHOST** and **USERPLANEHOST** MOs must be both turned on or off.

Figure 5-5 Configuration model of direct IPsec sharing among multiple operators



5.1.3 Static Blacklist & Whitelist

Static Blacklist

A security peer can be added to the static blacklist of a local end for direct IPsec tunnels, to prevent the local end from establishing a direct IPsec tunnel with the peer end. This function is applicable when a local gNodeB can establish an eXn interface with a peer gNodeB through the SeGW. The **ADD DIRECTIPSECBLACKLIST** command can be executed to add a static direct IPsec blacklist entry. You are advised to configure the static direct IPsec blacklist before enabling the direct IPsec function. If direct IPsec tunnels have been set up between the local and peer base stations before the blacklist is configured, you need to manually remove the **SECURITYPEER** MO before configuring a static direct IPsec blacklist.

Static Whitelist

Network segments can be added to the static whitelist of a local end for direct IPsec tunnels, to allow the local end to establish direct IPsec tunnels with only the peer ends on the network segments within the whitelist. This function is applicable when the gNodeB only needs to establish direct IPsec tunnels to the peer ends on some network segments. The **ADD DIRECTIPSECWHITELIST** command can be executed on a gNodeB to add a static direct IPsec whitelist entry.

 NOTE

- The static blacklist and static whitelist functions cannot both be configured.
- The static whitelist and direct IPsec self-removal functions cannot both be configured.

5.1.4 Self-Update

gNodeBs support self-update of direct IPsec, which is controlled by the **GEMODEL PARA.DIRIPSECUPDATESW** parameter. Assume that an eXn interface is functioning properly after self-update of direct IPsec is enabled. An eXn self-setup procedure is triggered if the eXn control plane (the Xn control plane in this version) is still functioning properly after any one of the following reconfigurations is performed. Then, the local base station sends information about direct IPsec to the peer base station. A direct IPsec self-update is performed if direct IPsec is enabled on the peer base station. The reconfigurations are as follows:

- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the **EPGROUP** MO of the eXn interface is changed from **DISABLE** to **ENABLE**, and the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is **DISABLE**.
- (Applied only to direct IPv4 IPsec self-setup) The endpoint group of the eXn interface is modified with a new **SECURITYHOST** MO referenced, and the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is **DISABLE**.
- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the eXn interface is **ENABLE**, and the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is changed from **ENABLE** to **DISABLE**.
- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the eXn interface is **ENABLE**, the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is **DISABLE**, and the **SECURITYHOST.LOCIKEIPV4** parameter value is changed.
- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the eXn interface is **ENABLE**, the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is **DISABLE**, and the **SECURITYHOST.LOCIKEIPV6** parameter value is changed.
- The **ADD GNBDUEXN** command is executed to add an eXn interface, or the **MOD GNBDUEXN** command is executed to modify the **GNBDUEXN.UpEpGroupId** parameter corresponding to an eXn interface.

There is a 3-minute configuration protection period for the self-update of direct IPsec. If any one of the preceding reconfigurations is executed, the configuration protection is triggered. During a configuration protection period, no direct IPsec self-update is triggered. After the configuration protection period ends, if the configuration is changed, a self-update of direct IPsec is triggered.

After an eXn interface between the base station and each neighboring base station is automatically set up, a 100s protection timer is automatically created. In direct IPsec scenarios, if a user-plane IP address is changed within the protection timer length, the eXn self-update fails. The procedure can be restarted only after the protection timer expires.

 NOTE

If the eXn link is normal but no direct IPsec tunnel is set up, you can set the **GEPMODELPARA.DIRIPSECUPDATESW** parameter to **ENABLE** to enable the eXn self-update to trigger self-setup of direct IPsec.

If the eXn control plane is normal and direct IPsec is configured on the local end, direct IPsec self-update is initiated on the user plane when a base station is reset.

5.1.5 Self-Removal

Self-removal of direct IPsec applies when direct IPsec is enabled on a gNodeB and its connected gNodeBs, but direct IPsec tunnels are unreachable to the connected gNodeBs due to transport network planning. If such a scenario does not exist, this function is not recommended. This function is controlled by the **GEPMODELPARA.DIRIPSECAUTODELSW** parameter.

When detecting a direct IPsec fault caused by IKE SA negotiation failure: the gNodeB does not report an alarm but removes the faulty link if the **GEPMODELPARA.DIRIPSECAUTODELTIMER** parameter is set to **0**; the gNodeB reports ALM-25891 IKE Negotiation Failure if the **GEPMODELPARA.DIRIPSECAUTODELTIMER** parameter is set to a non-zero value. This facilitates transmission fault locating. The gNodeB waits for a period specified by the **GEPMODELPARA.DIRIPSECAUTODELTIMER** parameter. If the fault persists after the timer expires, the gNodeB removes the direct IPsec tunnel and clears ALM-25891 IKE Negotiation Failure. It is recommended that the **GEPMODELPARA.DIRIPSECAUTODELTIMER** parameter value be set to one week or longer. If the configured period is excessively short, it is possible that the gNodeB has triggered direct IPsec self-removal and related alarms have been cleared before users troubleshoot transmission faults based on the alarms, or ALM-25891 IKE Negotiation Failure is reported again after it is cleared due to the check mechanism.

If gNodeB configuration is prohibited, or there is a large number of to-be-removed direct IPsec tunnels, the fault recovery time will be longer than the period specified by the **GEPMODELPARA.DIRIPSECAUTODELTIMER** parameter. This gap is dependent on the gNodeB configuration prohibition duration or the number of to-be-removed direct IPsec tunnels.

5.1.6 Application Restrictions

Direct IPv6 IPsec requires that the value of the **EPGROUP.IPV6VRFIDX** parameter differ from that of the **SECURITYHOST.VRFIDX** parameter.

When direct IPsec is configured for the eXn user plane, it is recommended that the NG and eXn interfaces reference different **USERPLANEHOST** MOs. This is because the **USERPLANEHOST** MO referenced by the NG interface requires the IPsec self-setup switch (**USERPLANEHOST.IPSEC SWITCH**) to be turned off but the **USERPLANEHOST** MO referenced by the eXn interface requires this switch to be turned on.

When direct IPsec is configured only for the eXn user plane and the configuration of the IPsec self-setup switch in the **USERPLANEHOST** MO changes, the configuration change takes effect only when eXn self-setup is performed.

Direct IPsec self-setup does not apply to IPv4-over-IPv6 IPsec or IPv6-over-IPv4 IPsec. IPv4-over-IPv6 IPsec indicates that IPv4 is used for the inner IP header and

IPv6 for the outer IP header. IPv6-over-IPv4 IPsec indicates that IPv6 is used for the inner IP header and IPv4 for the outer IP header. For details about IPv4-over-IPv6 IPsec and IPv6-over-IPv4 IPsec, see *IPsec*.

5.2 Network Analysis

5.2.1 Benefits

On a network secured using IPsec, SeGW networking results in significant delay over the eXn interface, whereas direct IPsec networking reduces the length of the transmission path, which reduces the delay over the eXn interface.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

IPsec ensures transmission security by encapsulating and encrypting IP packets. However, this reduces the transmission efficiency of service packets on the bearer network. For details, see *IPsec*.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010080	IPsec	NR0S0IPSEC00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Transmission configuration mode	GTRANSPARA.TRANS CFG MODE	<i>eXn Self-Management</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNS WITCH parameter) can be turned on only when the transmission mode is configured in new mode (by setting the GTRANSPARA.TRANS CFG MODE parameter to NEW).
Low-frequency TDD High-frequency TDD	Self-removal of direct IPsec	GEPMODEL PARA.DIRIPSE CAUTODELS W	<i>eXn Self-Management</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNS WITCH parameter) can be turned on only when the switch for self-removal of direct IPsec is turned on.
Low-frequency TDD High-frequency TDD	Security protocol	IPSECPROPOSAL.TRAN MODE	<i>IPsec</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNS WITCH parameter) can be turned on only when the IPSECPROPOSAL.TRAN MODE parameter is set to ESP .

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	IPsec configuration mode	IKECFG.IPSECCFGMODE	<i>eXn Self-Management</i>	In IPv4 scenarios, if IKECFG.IPSECCFGMODE is set to ETHINTERFACE_BASED , the multi-VPN switch (SECURITYTEMPLATE.MULTIVPNSWITCH) of the SECURITYTEMPLATE MO referenced by the SECURITYHOST MO for direct IPsec (with SECURITYHOST.SEGWSWITCH set to DISABLE) cannot be turned on.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator-specific eXn interface configuration	MO gNB DUExnOP	Multi-Operator Sharing	When operators are configured with different eXn MOs and they form alliances with each other, if the alliances are inconsistent between operators on neighboring base stations: • After self-update of direct IPsec is enabled, USERPLANEPEER MOs may be added to the EPGROUP MO. This occupies the USERPLANEPEER MO specifications of the endpoint group, including the specifications of the entire base station and a single endpoint group. • After self-update of direct IPsec is enabled, eXn self-update may fail and the base station reports ALM-25954 User Plane Fault.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator-specific eXn transmission path setup in separated mode	FLEX_EXNOP_SW option of the gNBInterfaceParam.InterfacePolicySwitch parameter	<i>Multi-Operator Sharing</i>	If (1) operator-specific eXn transmission path setup in separated mode is not enabled for all base stations that support coordination-based services, and (2) an operator is configured with separated and shared eXn interfaces on the local and peer base stations, respectively, the following rules apply: • After self-update of direct IPsec is enabled, the local and peer base stations may add user-plane peers to an endpoint group, consuming the specifications of user-plane peers of the endpoint group. Such specifications include the specifications of the entire base station and a single endpoint group. • After self-update of direct IPsec is

RAT	Function Name	Function Switch	Reference	Description
				enabled, eXn self-update may fail on the local and peer base stations. If this is the case, ALM-25954 User Plane Fault is reported.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Direct IPsec sharing among multiple operators over the eXn interface: Only the UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series main control boards and all NR-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Other functions: Only the UMPThb, UMPTha, UMPTg, UMPTga, and UMPTe series main control boards, and only the UBBPg, UBBPh, and UBBPfw series baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

There are no requirements for RF modules.

5.3.4 Others

The end-to-end MTU of eXn transmission links must be greater than or equal to 1500 bytes. This will reduce the fragmentation of coordination packets, which can affect coordination services.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Direct IPsec for eXn interface is based on eXn self-setup between gNodeBs. For details about the configurations for eXn self-setup, see [4.4.1.1 Data Preparation](#). This section describes only IPsec configurations.

The following table describes the parameters that must be set in a **GTRANSPARA** MO to configure the direct IPsec priority matching switch.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Priorities Match Switch	GTRANSPARA.DIRECTIP SECPRIMATCHSW	Set this parameter to ENABLE to deploy direct IPsec. This parameter applies only to direct IPv4 IPsec.

The following table describes the parameters that must be set in an **IKEPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
Proposal ID	IKEPROPOSAL.PROPID	Set this parameter based on the network plan.
Authentication Method	IKEPROPOSAL.AUTHME TH	Set this parameter to IKE_CERT_SIG if digital-certificate-based authentication is required. Set this parameter to the same value for all gNodeBs in the network.

The following table describes the parameters that must be set in an **IPSECPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
IPsec Proposal Name	IPSECPROPOSAL.PROP NAME	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Transform	IPSECPROPOSAL.TRAN MODE	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **SECURITYTEMPLATE** MO. This MO applies only to IPsec-enabled scenarios. When the following parameters are to be modified, the base station needs to adjust the negotiation status of all security links automatically created using the **SECURITYTEMPLATE** MO. The adjustment takes a long time. Therefore, you are advised to run the **MOD SECURITYTEMPLATE** command to modify parameters, wait for about 30 seconds, and then run other commands.

Parameter Name	Parameter ID	Setting Notes
Security Parameters Template ID	SECURITYTEMPLATE.SECURITYTEMPLATEID	Set this parameter based on the network plan.
SeGW Switch	SECURITYTEMPLATE.SEGWSWITCH	Set this parameter to DISABLE to enable gNodeBs to adopt direct IPsec negotiation.
IKE Proposal ID	SECURITYTEMPLATE.IKEPROPID	Set this parameter based on the network plan.
IPsec Proposal Name	SECURITYTEMPLATE.IPSECPROPNAME	Set this parameter based on the network plan.
IKE Version	SECURITYTEMPLATE.IKEVERSION	Set this parameter based on the network plan.
DPD Mode	SECURITYTEMPLATE.DPD	Set this parameter based on the network plan.
Certificate Source	SECURITYTEMPLATE.CERTSOURCE	Set this parameter based on the network plan.
IKEv2 Fragmentation Switch	SECURITYTEMPLATE.IKEV2FRGSW	Set this parameter based on the network plan.
Multiple VPN Switch	SECURITYTEMPLATE.MULTIVPNSWITCH	Set this parameter to ON when direct IPsec sharing among multiple operators over the eXn interface is enabled.
Narrow Down Switch	SECURITYTEMPLATE.NARROWDOWNSW	Set this parameter to ON when direct IPsec sharing among multiple operators over the eXn interface is enabled.

The following table describes the parameters that must be set in a **SECURITYHOST** MO.

Parameter Name	Parameter ID	Setting Notes
Security Host ID	SECURITYHOST.SECURITYHOSTID	Set this parameter based on the network plan.
SeGW Switch	SECURITYHOST.SEGWSWITCH	Set this parameter to DISABLE to enable gNodeBs to adopt direct IPsec negotiation.
IP Version	SECURITYHOST.IPVERSION	Set this parameter based on the network plan.
Local IKE IP Address	SECURITYHOST.LOCIKEIPV4	Set this parameter based on the network plan.
Local IKE IPv6 Address	SECURITYHOST.LOCIKEIPV6	Set this parameter based on the network plan.
Security Parameters Template ID	SECURITYHOST.SECURITYTEMPLATEID	Set this parameter based on the network plan.
VRF Index	SECURITYHOST.VRFIDX	Set this parameter based on the network plan. The value of this parameter must be different from the values of SCTPHOST.VRFIDX and USERPLANEHOST.VRFIDX . This parameter applies only to direct IPv6 IPsec.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO to configure the local gNodeB information of the eXn-U interface.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPHOSTID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	USERPLANEHOST.VRFID_{DX}	Set this parameter based on the network plan. The value of this parameter must be the same as that of the SCTPHOST.VRFID_{DX} parameter but different from that of the SECURITYHOST.VRFID_{DX} parameter. This parameter applies only to direct IPv6 IPsec.
IP Version	USERPLANEHOST.IPVERSION	Set the IP version of the local end to the same as that of the peer end.
Local IP Address	USERPLANEHOST.LOCIPV4	If USERPLANEHOST.IPVERSION is set to IPv4 , configure a local user-plane IPv4 address.
Local IPv6 Address	USERPLANEHOST.LOCIPV6	If USERPLANEHOST.IPVERSION is set to IPv6 , configure a local user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEHOST.IPSEC SWITCH	Set this parameter to ENABLE to deploy direct IPsec.
Security Host ID	USERPLANEHOST.SECHOSTID	Set this parameter based on the network plan.
User Label	USERPLANEHOST.USER LABEL	Set this parameter based on the network plan.

For the parameters that must be set in an **EPGROUP** MO to configure an endpoint group for the eXn user plane, see [4.4.1.1 Data Preparation](#).

For the parameters that must be set to configure the reference relationship between the **gNBDUExn** and **EPGROUP** MOs, see [4.4.1.1 Data Preparation](#).

(Optional) The following table describes the parameters that must be set in a **GEPMODEL PARA** MO to configure self-removal of direct IPsec.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Automatic Deletion Switch	GEPMODEL PARA.DIRIPSECAUTODELSW	Set this parameter based on the network plan.
Direct IPsec Automatic Deletion Timer	GEPMODEL PARA.DIRIPSECAUTODELTIMER	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to configure self-update of direct IPsec.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Update Switch	GEPMODELPARA.DIRIPSECUPDATESW	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **DIRECTIPSECBLACKLIST** MO.

Parameter Name	Parameter ID	Setting Notes
ID of Blacklist for Direct IPsec Tunnel	DIRECTIPSECBLACKLIST.DIRECTIPSECBLACKLISTID	Set this parameter based on the network plan.
IP Version	DIRECTIPSECBLACKLIST.IPVERSION	Set this parameter based on the network plan.
Security Peer IPv4 Address	DIRECTIPSECBLACKLIST.SECURITYPEERIPV4	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECBLACKLIST.IPVERSION parameter is set to IPV4 .
Security Peer IPv6 Address	DIRECTIPSECBLACKLIST.SECURITYPEERIPV6	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECBLACKLIST.IPVERSION parameter is set to IPV6 .

The following table describes the parameters that must be set in a **DIRECTIPSECWHITELIST** MO.

Parameter Name	Parameter ID	Setting Notes
ID of Whitelist for Direct IPsec Tunnel	DIRECTIPSECWHITELIST.DIRECTIPSECWHITELISTID	Set this parameter based on the network plan.
IP Version	DIRECTIPSECWHITELIST.IPVERSION	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Security Peer IPv4 Address	DIRECTIPSECWHITELIST.SECURITYPEERIPV4	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV4 .
Mask	DIRECTIPSECWHITELIST.MASK	Set this parameter based on the network plan.
Security Peer IPv6 Address	DIRECTIPSECWHITELIST.SECURITYPEERIPV6	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV6 .
IPv6 Address Prefix Length	DIRECTIPSECWHITELIST.IPV6PFXLEN	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV6 .

(Optional) Configure the **VRF** MO to configure direct IPsec sharing among multiple operators over the eXn interface.

Parameter Name	Parameter ID	Setting Notes
User Label	VRF.USERLABEL	Set this parameter based on the network plan.
IPSec VPN Switch	VRF.IPSECVPNBW	Set this parameter to ENABLE when direct IPsec sharing among multiple operators over the eXn interface is enabled.
IPSec VPN ID	VRF.IPSECVPNID	Set this parameter based on the network plan. Operators must negotiate with each other and ensure that the parameter setting of each operator is consistent across all of its base stations.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configuring direct IPsec for eXn interface:

```
//Configuring direct IPsec
//Setting the direct IPsec matching prioritization with DIRECTIPSECPRIMATCHSW set to ENABLE. This
parameter applies only to direct IPv4 IPsec.
SET GTRANSPara: DIRECTIPSECPRIMATCHSW=ENABLE;
//Adding an IKE security proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", TRANMODE=ESP, ESPENCALG=AES128;
//Adding a security parameter template for direct IPv4 IPsec self-setup with SeGW Switch set to DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="*****", DPD=PERIODIC, LTCFG=LOCAL;
//Adding a security parameter template for direct IPv6 IPsec self-setup with SeGW Switch set to DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="*****", DPD=PERIODIC, LTCFG=LOCAL;
//Adding a security host (IPv4) with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4,
LOCIKEIPV4="7.7.4.146", SECURITYTEMPLATEID=0;
//Adding a security host (IPv6) with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv6,
LOCIKEIPV6="2000:0:0:0:0:0:1", SECURITYTEMPLATEID=1, VRFIDX=2;
```

Configuring an eXn interface and enabling direct IPsec (IPv4 used as an example):

```
//Turning on the switch for coordination data transmission over the backhaul port
MOD BBP: CN=0, SRN=0, SN=2, BBWS=NR-1, EXFUNCSW=COORD_OUTPUT_BACKHAUL_PT-1;
//Adding an endpoint group for the eXn interface
ADD EPGROUP: EPGROUPID=1, VRFIDX=0;
//Adding a user-plane host for the eXn interface and enabling direct IPsec
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=0, IPVERSION=IPv4, LOCIPV4="7.7.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the eXn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding an eXn object
ADD GNBDUEXN: gNBDuExnId=0, UpEpGroupId=1;
```

Configuring an eXn interface and enabling direct IPsec (IPv6 used as an example):

```
//Turning on the switch for coordination data transmission over the backhaul port
MOD BBP: CN=0, SRN=0, SN=2, BBWS=NR-1, EXFUNCSW=COORD_OUTPUT_BACKHAUL_PT-1;
//Adding an endpoint group for the eXn interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1;
//Adding a user-plane host for the eXn interface and enabling direct IPsec
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, LOCIPV6="2000:0:0:0:0:0:1:1",
IPSECSWITCH=ENABLE, SECHOSTID=1;
//Adding the eXn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding an eXn object
ADD GNBDUEXN: gNBDuExnId=0, UpEpGroupId=1;
```

Self-update and self-removal of direct IPsec for eXn interface:

```
//Enabling self-update of direct IPsec for eXn interface
SET GEPMODELpara: DIRIPSECUPDATESW=ENABLE;
//Enabling self-removal of direct IPsec for eXn interface
SET GEPMODELpara: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10080;
```

Blacklist and whitelist of direct IPsec for eXn interface:

```
//Adding a direct IPsec blacklist entry for the eXn interface (IPv4 used as an example)
ADD DIRECTIPSECBLACKLIST: DIRECTIPSECBLACKLISTID=0, IPVERSION=IPV4,
SECURITYPEERIPV4="7.7.4.154";
//Adding a direct IPsec blacklist entry for the eXn interface (IPv6 used as an example)
ADD DIRECTIPSECBLACKLIST: DIRECTIPSECBLACKLISTID=0, IPVERSION=IPV6,
SECURITYPEERIPV6="2000:0:0:0:0:1:2";
//Adding a direct IPsec whitelist entry for the eXn interface (IPv4 used as an example)
ADD DIRECTIPSECWHITELIST: DIRECTIPSECWHITELISTID=0, IPVERSION=IPV4,
SECURITYPEERIPV4="192.168.0.0", MASK="255.255.255.0";
//Adding a direct IPsec whitelist entry for the eXn interface (IPv6 used as an example)
ADD DIRECTIPSECWHITELIST: DIRECTIPSECWHITELISTID=0, IPVERSION=IPV6,
SECURITYPEERIPV6="2000:0:0:0:0:1:2", IPV6PFXLEN=64;
```

Configuring direct IPsec sharing among multiple operators over the eXn interface (using two operators as an example):

- Configuring for a newly deployed base station with IPv4 used as an example

```
//Configuring inner VRFs for two operators
ADD VRF: VRFID=1, USERLABEL="OP1", IPSECVPNNSW=ENABLE, IPSECVPNID=1;
ADD VRF: VRFID=2, USERLABEL="OP2", IPSECVPNNSW=ENABLE, IPSECVPNID=2;
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANSMODE=ESP, ESPENCALG=AES128;
SET GEPMODELpara: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
NARROWDOWNNSW=ON, IKEV2FRGSW=OFF, MULTIVPNNSW=ON;
//Configuring the security host information and referencing the preceding security parameter
template (VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPV4, VRFID=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Adding user-plane hosts for the two operators
ADD USERPLANEHOST: UPHOSTID=1, VRFID=1, IPVERSION=IPV4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
ADD USERPLANEHOST: UPHOSTID=2, VRFID=2, IPVERSION=IPV4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

- Configuring for an additional operator B with IPv4 used as an example

```
//Modifying the user-plane host for operator A and disabling direct IPsec
MOD USERPLANEHOST: UPHOSTID=1, VRFID=1, IPVERSION=IPV4, LOCIPV4="1.1.1.1",
IPSECSWITCH=DISABLE, USERLABEL="OP1";
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANSMODE=ESP, ESPENCALG=AES128;
SET GEPMODELpara: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
NARROWDOWNNSW=ON, IKEV2FRGSW=OFF, MULTIVPNNSW=ON;
//Configuring the security host information and referencing the preceding security parameter
template (VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPV4, VRFID=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Configuring inner VRFs for two operators
MOD VRF: VRFID=1, USERLABEL="OP1", IPSECVPNNSW=ENABLE, IPSECVPNID=1;
ADD VRF: VRFID=2, USERLABEL="OP2", IPSECVPNNSW=ENABLE, IPSECVPNID=2;
//Modifying the user-plane host for operator A and enabling direct IPsec again
MOD USERPLANEHOST: UPHOSTID=1, VRFID=1, IPVERSION=IPV4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
//Adding a user-plane host for operator B
ADD USERPLANEHOST: UPHOSTID=2, VRFID=2, IPVERSION=IPV4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

Optimization Command Examples

N/A

Deactivation Command Examples

Disabling direct IPsec for eXn interface

```
//Setting IPsec Auto Configure Switch in the USERPLANEHOST MO to DISABLE  
MOD USERPLANEHOST: UPHOSTID=1, IPSECSWITCH=DISABLE;
```

Disabling direct IPsec sharing among multiple operators over the eXn interface
(using two operators as an example):

```
MOD USERPLANEHOST: UPHOSTID=1, IPSECSWITCH=DISABLE;  
MOD USERPLANEHOST: UPHOSTID=2, IPSECSWITCH=DISABLE;  
MOD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, IKEVERSION=IKE_V2,  
MULTIVPNSWITCH=OFF;  
MOD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPSW=DISABLE;  
MOD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPSW=DISABLE;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Direct IPsec for the eXn-U Interface

Step 1 Run the **DSP IKESA** command to check the IKE SA negotiation results.

Expected result: The IDs of ACL rules generated during direct IPsec self-setup for the eXn interface range from 80000 to 84999. If the command output includes an IKE SA with the rule ID in this range, direct IPsec negotiation between the local and peer base stations has succeeded, and the IKE SA between the base station pair is normal.

Step 2 Run the **LST IPSECPOLICY** command to obtain **Policy Group Name** and **IPSec Sequence No.** corresponding to the ACL ID queried in the previous step.

Step 3 Run the **DSP IPSECSA** command to display the results of IPsec SA negotiation based on **IPSec Policy Group Name** and **IPSec Sequence No.**

Expected result: The IPsec SA information is displayed, which indicates that direct IPsec negotiation between the local and peer base stations has succeeded, and the IPsec SA between the base stations is normal.

----End

Self-Update of Direct IPsec

Start an Xn interface trace task three minutes after self-update of direct IPsec is enabled (the eXn-C setup currently requires the Xn-C). If an XNAP_PRIVATE_MSG message exists, this function has been activated.

Self-Removal of Direct IPsec

If ALM-25891 IKE Negotiation Failure is reported, the alarm is cleared automatically after the **Direct IPsec Automatic Deletion Timer** expires (if this timer length is not 0). Then, run the **LST IKEPEER** command to view the configurations of the IKE peer.

Expected result: The information about the IKE peer corresponding to the peer IP address is unavailable. This indicates that this function has taken effect.

5.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *IP Active Performance Measurement*
- *IP Performance Monitor*
- *Common Clock*
- *IPsec*
- *Equipment Security*
- *X2 and S1 Self-Management in NSA Networking*
- *NG and Xn Self-Management*
- *Multi-Operator Sharing*
- *License Control Item Lists*
- *Cloud BB Overview in eRAN Feature Documentation*
- *USU3910-based Multi-BBU Interconnection in eRAN Feature Documentation*